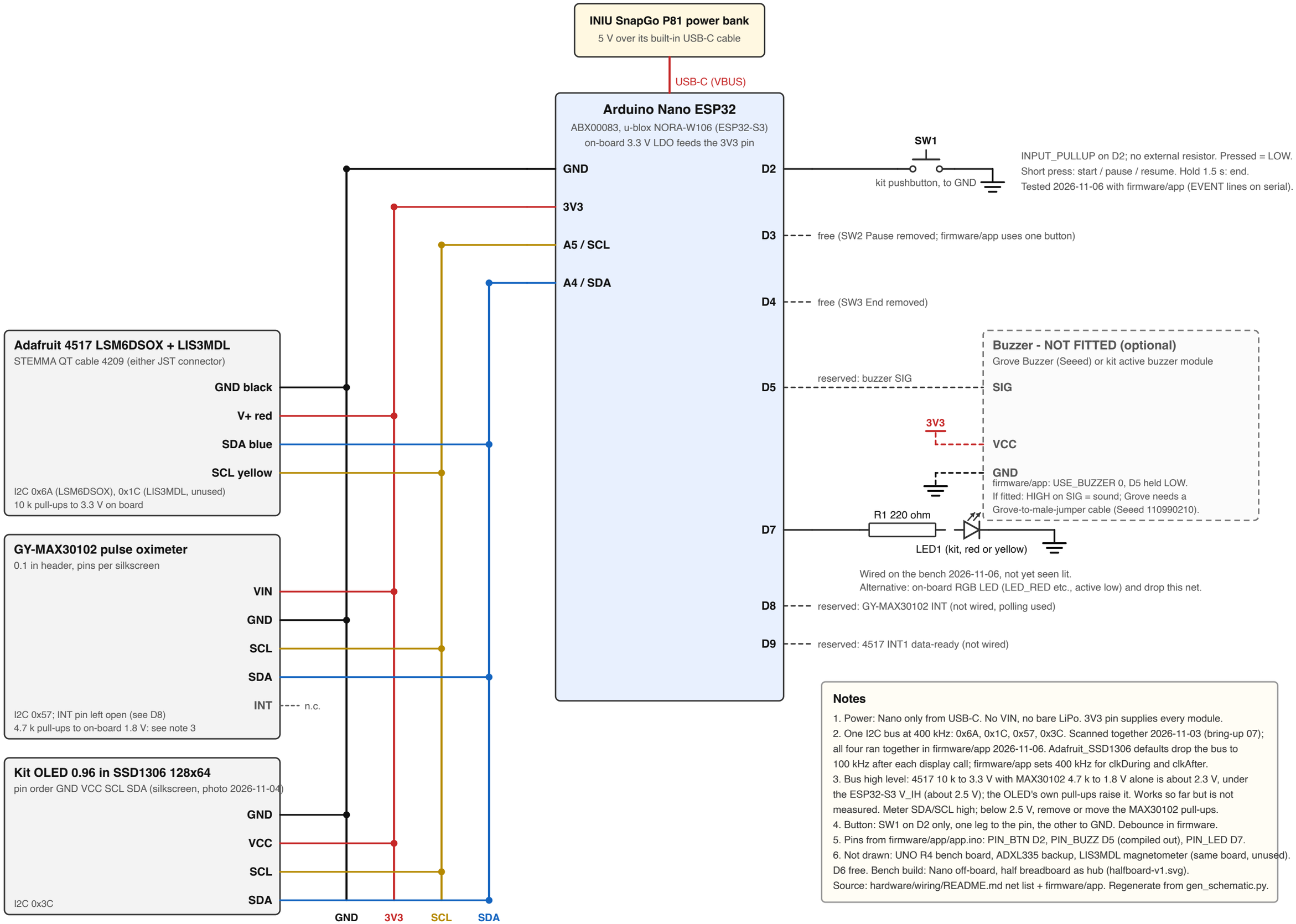
Smart dumbbell - schematic v1.2 (2026-11-08)

Modules as photographed 2026-11-04. I2C section: all four devices scanned 2026-11-03 (0x1C 0x3C 0x57 0x6A) and run together by firmware/app on 2026-11-06. SW1 tested 2026-11-06; LED wired, not yet observed.
Pin names are the Arduino names printed on the Nano ESP32 header. All logic is 3.3 V. A dot is a junction; a crossing without a dot is not a connection. Dashed = not fitted. Change from v1: one button (SW1), buzzer not fitted.



Notes

1. Power: Nano only from USB-C. No VIN, no bare LiPo. 3V3 pin supplies every module.
2. One I2C bus at 400 kHz: 0x6A, 0x1C, 0x57, 0x3C. Scanned together 2026-11-03 (bring-up 07); all four ran together in firmware/app 2026-11-06. Adafruit_SSD1306 defaults drop the bus to 100 kHz after each display call; firmware/app sets 400 kHz for clkDuring and clkAfter.
3. Bus high level: 4517 10 k to 3.3 V with MAX30102 4.7 k to 1.8 V alone is about 2.3 V, under the ESP32-S3 V_IH (about 2.5 V); the OLED's own pull-ups raise it. Works so far but is not measured. Meter SDA/SCL high; below 2.5 V, remove or move the MAX30102 pull-ups.
4. Button: SW1 on D2 only, one leg to the pin, the other to GND. Debounce in firmware.
5. Pins from firmware/app/app.ino: PIN_BTN D2, PIN_BUZZ D5 (compiled out), PIN_LED D7.
6. Not drawn: UNO R4 bench board, ADXL335 backup, LIS3MDL magnetometer (same board, unused). D6 free. Bench build: Nano off-board, half breadboard as hub (halfboard-v1.svg).

Source: hardware/wiring/README.md net list + firmware/app. Regenerate from gen_schematic.py.